

ASSIGNMENT I

How one atom becomes thermal inside a large isolated quantum system

Following the idea that I have sketched today, let us follow the steps below.

Consider N two-level systems. Each site has basis states $|0\rangle, |1\rangle$, with energies

$$\epsilon_0 = 0, \quad \epsilon_1 = \Delta.$$

Next, we split the system into:

- one distinguished atom S ,
- the remaining $N - 1$ atoms, that we call the bath B .

The Hilbert space factorizes as

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B.$$

The subsystem S has dimension 2, while the bath has dimension 2^{N-1} .

Assume the total energy is fixed to E .

Problem 1: Counting bath states

Suppose the bath consists of $N - 1$ noninteracting two-level systems. If exactly m bath atoms are excited, what is the bath energy $E_B(m)$? How many bath states have that energy?

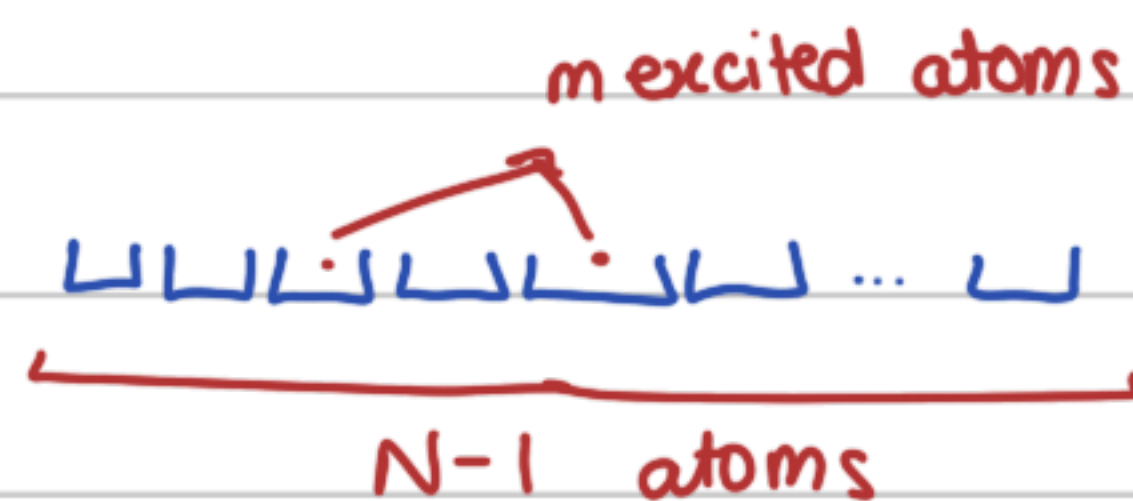
Solution

An excited states has energy $\epsilon_1 = \Delta$.

m excited atoms $\Rightarrow E_B(m) = m\Delta$

The number of states with m excited atoms from a total of $N-1$ atoms.

$$\Omega_B(m) = \binom{N-1}{m}$$



Problem 2: Fixing the total energy

Suppose the total energy of the system is fixed to

$$E = n\Delta.$$

If the subsystem S is in state $|0\rangle$, what must the bath energy be?

If S is in state $|1\rangle$, what must the bath energy be?

Solution

Case S in state $|0\rangle$:

Then, $\epsilon_s = 0$.

Therefore, all energy is in the bath.

$$E_B = n\Delta$$

Case in state $|1\rangle$:

Then, $\epsilon_s = \Delta$.

$$\begin{aligned} \text{Therefore, } E_B &= n\Delta - \Delta \\ &= (n-1)\Delta \end{aligned}$$

Problem 3: Probabilities from counting

Assume that within the total energy shell $E = n\Delta$, all allowed basis states are equally likely.

Show that the probability that the subsystem is in $|0\rangle$ is proportional to $\Omega_B(n)$, and that the probability that it is in $|1\rangle$ is proportional to $\Omega_B(n-1)$.

Solution

The probability to find system s is proportional to the number of states of s . i.e.

$$P(s) = \frac{\Omega(s)}{\Omega} \Rightarrow P(s) \propto \Omega(s)$$

Case $S = |0\rangle$:

The bath has n excited atoms.

$$\begin{aligned} \therefore P(|0\rangle) &\propto \Omega_B(n) \\ &= \binom{N-1}{n} \end{aligned}$$

Case $S = |1\rangle$:

The bath has $n-1$ excited atoms.

$$\begin{aligned} \therefore P(|1\rangle) &\propto \Omega_B(n-1) \\ &= \binom{N-1}{n-1} \end{aligned}$$

Problem 4: Ratio of probabilities

Show that

$$\frac{P(1)}{P(0)} = \frac{n}{N-n}.$$

This shows that the subsystem occupation is determined by the bath state count.

Solution

$$\frac{P(1)}{P(0)} = \frac{\Omega_B^{(n-1)}}{\Omega_B^{(n)}} = \frac{\binom{N-1}{n-1}}{\binom{N-1}{n}} = \frac{\frac{(N-1)!}{(n-1)!(N-1-n+1)!}}{\frac{(N-1)!}{n!(N-1-n)!}} = \frac{n}{N-n}$$

Problem 5: Emergence of the Boltzmann factor

Show that the ratio

$$\frac{P(1)}{P(0)}$$

can be written in the form

$$e^{-\beta\Delta}$$

for a suitable inverse temperature β . Find β .

Solution

$$\text{Let } \frac{P(1)}{P(0)} = e^{-\beta\Delta} \text{ for some } \beta.$$

$$\text{Then, } e^{-\beta\Delta} = \frac{n}{N-n}$$

$$-\beta\Delta = \ln\left(\frac{n}{N-n}\right)$$

$$\beta = \frac{1}{\Delta} \ln\left(\frac{N-n}{n}\right)$$

Problem 6: Reduced density matrix of the subsystem

Using the results of Problem 5, show that the subsystem density matrix has the thermal form

$$\rho_S = \frac{e^{-\beta H_S}}{Z}, \quad H_S = \Delta |1\rangle \langle 1|.$$

In other words, construct the density matrix step by step to show that it has the thermal (Gibbs) form.

Solution

Our problem is classical i.e. we are not considering super positions of states $|0\rangle$ & $|1\rangle$.

Therefore, our density matrix is a mixture of states $|0\rangle$ & $|1\rangle$.

$$\rho_S = P(0) |0\rangle \langle 0| + P(1) |1\rangle \langle 1|$$

From Problem 5,

$$\frac{P(1)}{P(0)} = e^{-\beta \Delta} \Rightarrow P(0) = \frac{1}{Z} \cdot P(1) = \frac{e^{-\beta \Delta}}{Z}$$

Let's now investigate the Boltzmann factor $e^{-\beta H_S}$.

$$\begin{aligned} H_S &= \epsilon_0 |0\rangle \langle 0| + \epsilon_1 |1\rangle \langle 1| \\ &= \Delta |1\rangle \langle 1| \end{aligned}$$

$$\begin{aligned} e^{-\beta H_S} &= e^{-\beta \epsilon_0} |0\rangle \langle 0| + e^{-\beta \epsilon_1} |1\rangle \langle 1| \\ &= |0\rangle \langle 0| + e^{-\beta \Delta} |1\rangle \langle 1| \end{aligned}$$

Comparing with ρ_S

$$\rho_S = \frac{1}{Z} |0\rangle \langle 0| + \frac{e^{-\beta \Delta}}{Z} |1\rangle \langle 1| = \frac{e^{-\beta H_S}}{Z}$$

Problem 7: General entropy argument

Show that generally

$$\Omega_B(E - \epsilon) \approx \Omega_B(E) e^{-\beta \epsilon}, \quad \beta = \frac{\partial S_B}{\partial E},$$

leading to the conclusion

$$P(\epsilon) \propto \Omega_B(E - \epsilon) \rightarrow e^{-\beta \epsilon}.$$

Our derivation shows that:

- thermal weights come from bath states,
- temperature comes from the entropy slope,
- Gibbs state emerges from normalization and ratios.

Thermal equilibrium = combinatorics of a large environment.

- no randomness assumed,
- no Gibbs state assumed,
- no ensemble postulate needed.

Solution

Suppose that our small subsystem has $E_s = \epsilon$ and thus our bath has $E_B = E - \epsilon$

From Problem 3 &

$$P(\epsilon) \propto \Omega_B(E - \epsilon)$$

Now relating the entropy with number of states $S(E) = k \ln \Omega(E)$:

$$\Omega_B(E - \epsilon) = \exp \left\{ \frac{S(E - \epsilon)}{k} \right\}$$

$$\approx \exp \left[\left(S(E) - \epsilon \frac{dS}{dE} \right) / k \right] \quad \text{since } \epsilon \ll E, \text{ we can do the linear approx.}$$

$$\approx \exp \left\{ \frac{S(E)}{k} \right\} \cdot \exp \left\{ -\epsilon \left(\frac{1}{k} \frac{dS}{dE} \right) \right\}$$

$$\approx \Omega_B(E) e^{-\beta \epsilon} \quad \beta = \frac{1}{k} \frac{dS}{dE}$$

Therefore, $P(\epsilon) \propto e^{-\beta \epsilon}$ $\square$